Evaluating YOLOv11m for Microplastic Morphology Detection & Classification and Developing a Web Application

Background

What are Microplastics?

- Plastic particles less than 5 mm in size
- Harmful to the environment — can absorb and release pollutants
- Current detection methods are slow or lack polymer type identification

Morphology types:

Fiber	Film	Pellet
Fragment	Foam	

Polymer types:

Polyethylene	Polypropylene	Polystyrene
Polyurethane	PVC	PET

Past Research

Akkajit et al. (2024) demonstrated YOLOv8x achieves high precision in microplastic morphology detection from microscope imagery.

Swinnerton et al. (2024) confirmed microplastics can be generated from large household plastics using a grinder and scissors.

YOLO models use deep learning to detect and classify objects within images in a single forward pass through a neural network.

Research Focus

How accurate is YOLOv11m when trained on a custom augmented dataset in detecting microplastic morphologies — and can a web application accurately identify both the morphology and polymer type of microplastics within uploaded microscope images?

References

[illegible]

Methodology

1 Plastic Collection

~10 household plastics collected: pillow filling (fibers), broken toys (fragments), plastic bags (films), microspheres (pellets), Styrofoam (foam).

2 Microplastic Creation

Plastics washed, cut with scissors, ground in spice grinder with ice to prevent melting. Stored in labeled glass beakers. PPE: N95 mask, gloves, goggles.

3 FTIR Verification

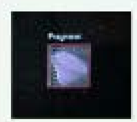
Each sample verified via ATR-FTIR spectrometer. Absorbance spectra matched to reference library. Pink plastic → polypropylene at 91.61% match.

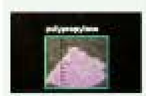
4 Imaging

Digital stereo microscope at 40× magnification. 1–4 particles per petri dish. ~230 base images captured and transferred to computer for annotation.

5 Labelling & Annotation

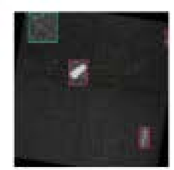
Two Roboflow projects created: morphology type and polymer type. Bounding boxes drawn manually around each particle with appropriate class labels.





6 Augmentation

Flip (H & V), Rotate ±25°, Zoom 20%, Noise 5%. Dataset expanded from ~230 to ~680 images. Split: 70% train / 20% validation / 10% test.



7 Model Training & Eval.

YOLOv11m trained for 100 epochs in Google Colab via Ultralytics. Two separate models: morphology type and polymer type. Evaluated on mAP@0.5, precision, recall, and F1.

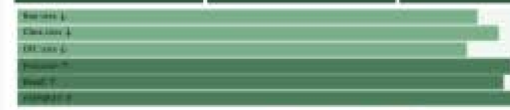
8 Web Application

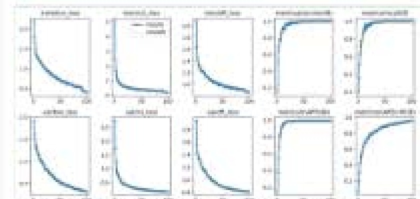
Dual YOLO detection app built in Streamlit. Accepts microscope image → runs both models → returns morphology and polymer type as JSON output.

Results

Morphology Classification Model

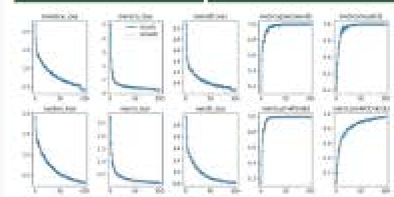
0.9991	0.9972	0.9981





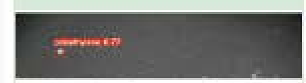
Polymer Type Classification Model

0.999	0.998	0.998

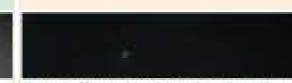


Web Application Results

Dual detection Streamlit app built successfully. Input = microscope image. Output = morphology + polymer type.



Polyethylene at 0.77 confidence. Independently verified correct.



Empty array [ ] returned. Low-light conditions diverged from training data.

Discussion & Conclusion

Key Findings

- High precision, recall, and F1 confirm strong model performance
- Performance gap between metrics and web app suggests possible overfitting
- Augmentation and 70/20/10 data split mitigated overfitting risk

Future Work

- Expand dataset with GAN-generated synthetic images
- Evaluate web app across varied imaging conditions
- Obtain more verified samples for comprehensive testing

Overall

This study demonstrates the viability of YOLOv11m for automated microplastic detection and classification. The Streamlit web application represents a meaningful step toward accessible AI-assisted environmental monitoring tools.